

CONCEPT MAP 1 : DECISION SUPPORT SYSTEMS (DSS)

1. TYPES OF DSS

- Data-driven DSS
Uses data and statistical analysis to support decisions.
- Model-driven DSS
Uses mathematical, financial or simulation models.
- Knowledge-driven DSS
Uses expert knowledge, rules and experience.
- Document-driven DSS
Manages, retrieves and analyzes unstructured documents.
- Communication-driven DSS
Supports group decision-making through communication tools.

DEFINITION

Decision Support Systems (DSS) are computer-based systems that help managers and decision makers analyze data, evaluate alternatives and make better decisions.

DECISION SUPPORT SYSTEM (DSS)

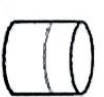
2. FLOW : DATA → ANALYSIS → DECISION

DATA
Collection of raw data from various sources.

ANALYSIS
Processing data using models, techniques and tools.

DECISION
Evaluate results and choose the best alternative.

CORE COMPONENTS



DATA
Raw facts and figures collected from internal or external sources.



MODEL
Mathematical, statistical or logical representations used to analyze the data.



USER INTERFACE
Allows users to interact with the system, input data and view results.

3. ROLE IN BUSINESS DECISION-MAKING

- Helps in analyzing complex and large amounts of data.
- Provides accurate and timely information.
- Improves quality of decisions.
- Reduces risks and uncertainties.
- Supports strategic planning and forecasting.
- Enhances efficiency and performance.

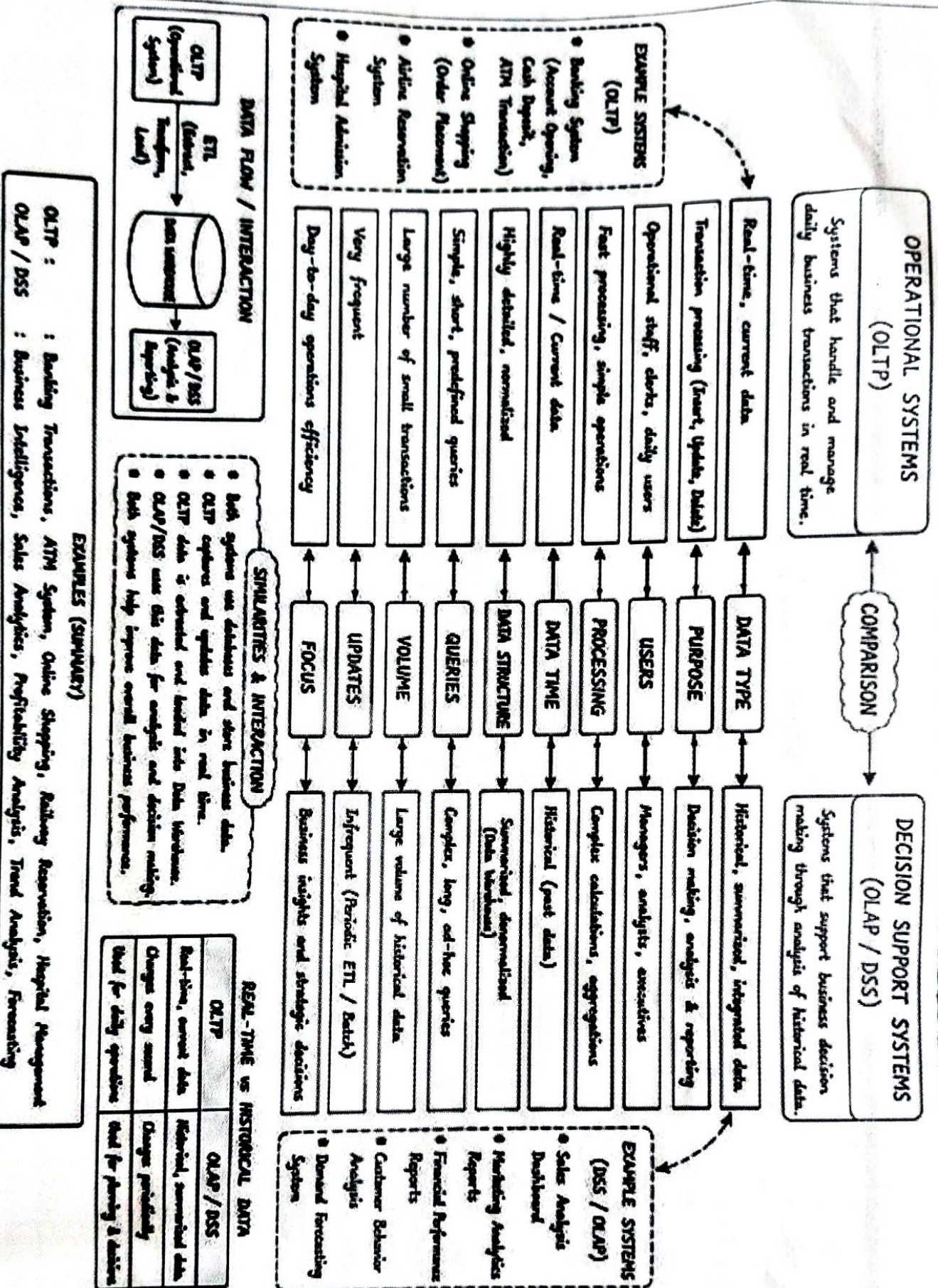
4. REAL-WORLD EXAMPLE LINKAGE

- Banking → Loan approval, credit scoring
- Healthcare → Patient diagnosis, treatment planning
- Retail → Sales forecasting, market basket analysis
- Manufacturing → Inventory planning, demand forecasting
- Education → Student performance analysis, placement prediction

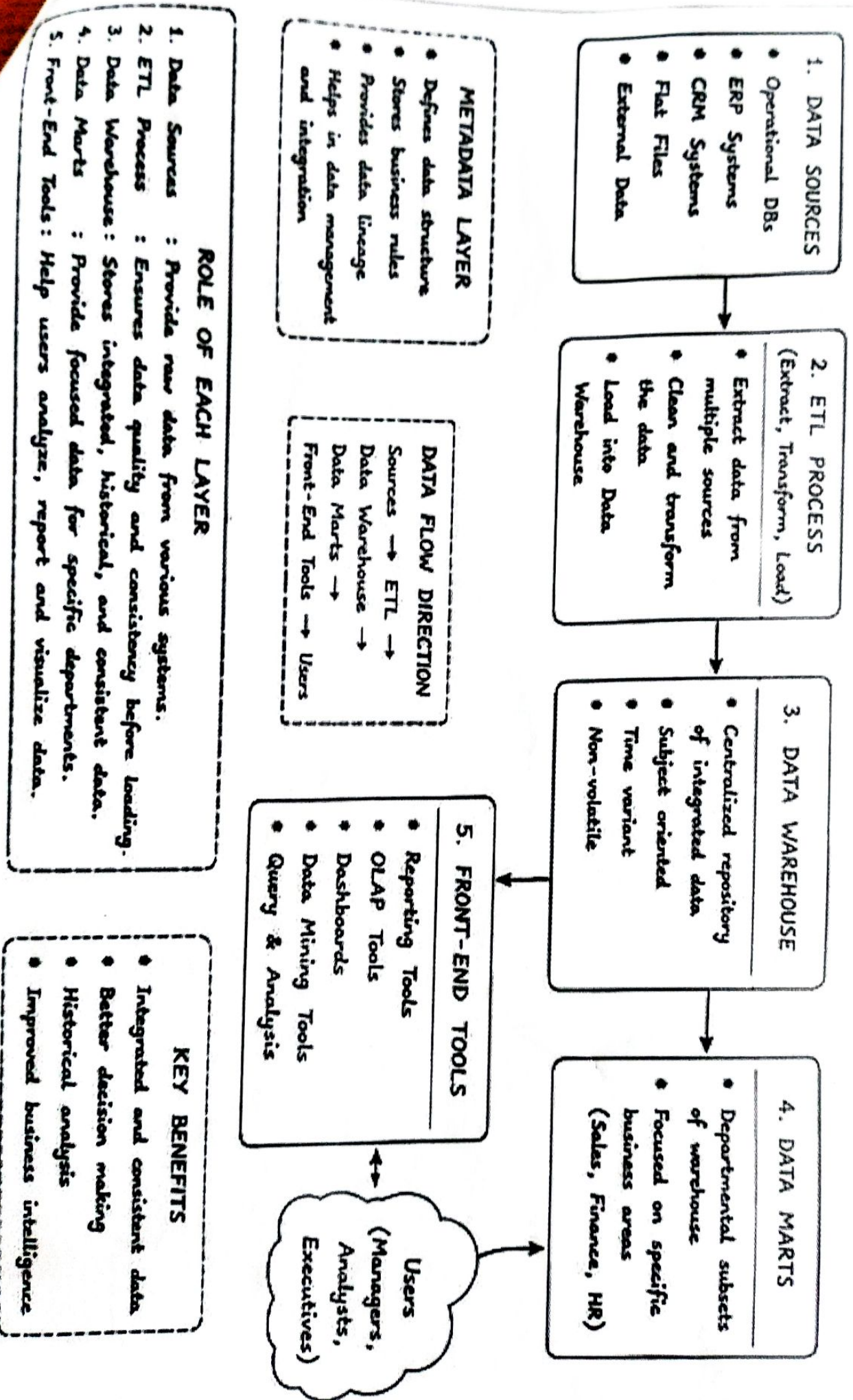
SUMMARY

DSS integrates data, models and user interface to analyze information, evaluate alternatives and support managers in making better, faster and more reliable decisions.

CONCEPT MAP 2 : OPERATIONAL SYSTEMS (OLTP) vs DSS (OLAP)

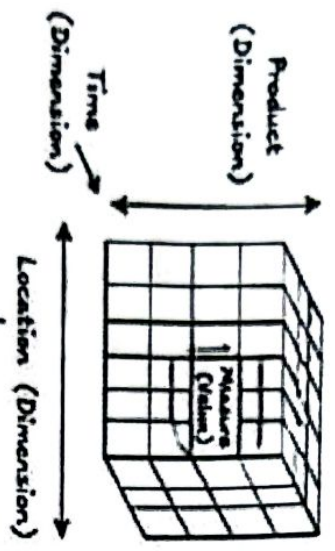


CONCEPT MAP 3 : ARCHITECTURE OF DATA WAREHOUSE



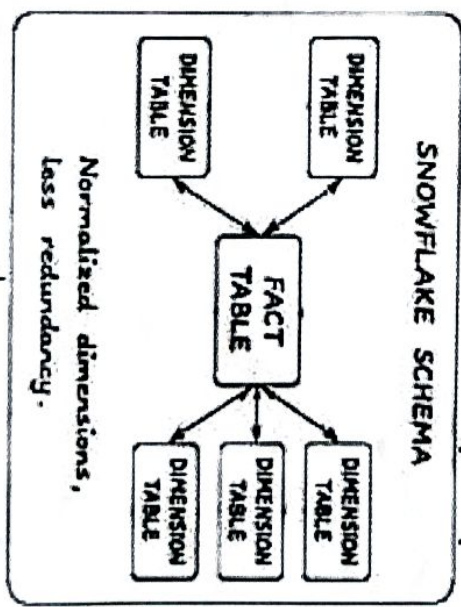
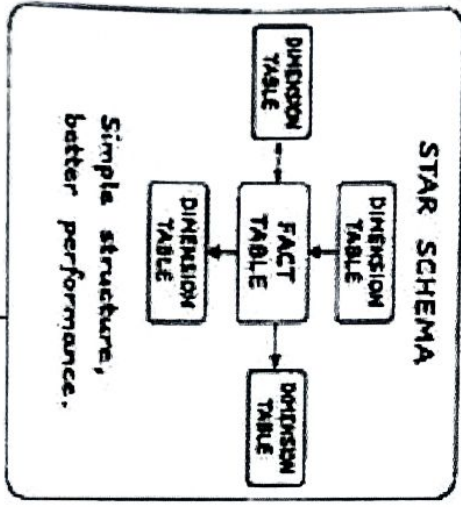
CONCEPT MAP 4 : OLAP, DATA CUBES & DIMENSIONAL MODELLING

1. DATA CUBE
Multidimensional structure used for data analysis.



- 2. OLAP OPERATIONS**
- **Slice** : Fix one dimension.
 - **Dice** : Select a sub-cube.
 - **Roll-up** : Summarize (higher level).
 - **Drill-down** : More detail (lower level).

3. DIMENSIONAL MODELLING
Organizes data for efficient analysis.



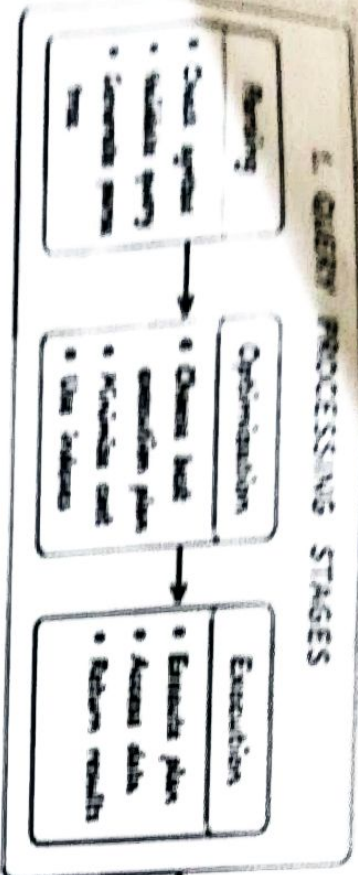
- 4. FACT TABLE**
- Contains measures (numeric data)
 - Foreign keys to dimension tables
 - Large table of transactional data

- 5. DIMENSION TABLE**
- Contains descriptive attributes
 - Provides context to facts
 - Smaller tables

- 6. ANALYTICAL USAGE**
- Trend Analysis
 - Performance Measurement
 - Forecasting
 - Market Basket Analysis
 - Customer Segmentation
 - Business Intelligence

CONCEPT MAP 5 : QUERY PROCESSING & BI TOOLS (KNIME)

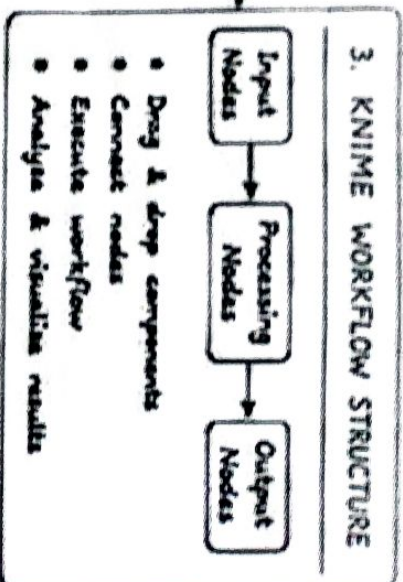
1. QUERY PROCESSING STAGES



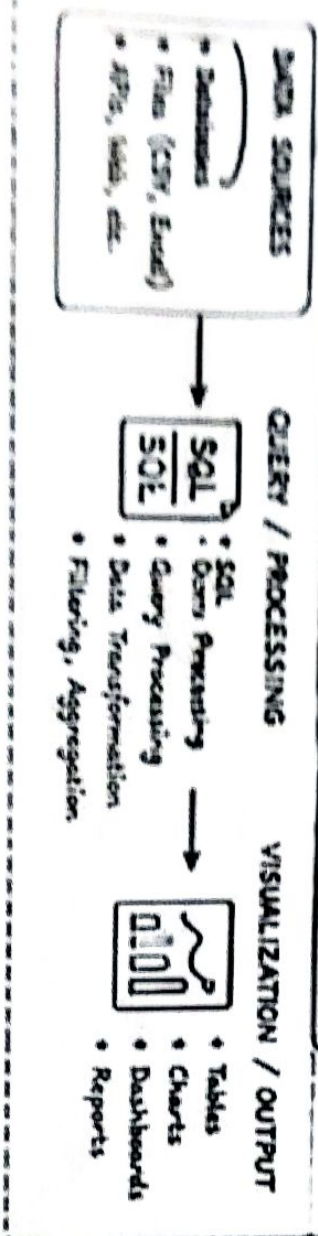
2. ROLE OF BI TOOLS

- Connect to multiple data sources
- Perform data transformation
- Enable analysis and visualization
- Provide interactive dashboards
- Support OLAP & reporting

3. KNIME WORKFLOW STRUCTURE



4. DATA → QUERY → VISUALIZATION PIPELINE



5. OLAP APPLICATION INTEGRATION

- OLAP operations on data cubes (Slice, Dice, Roll-up, Drill-down)
- Multi-dimensional analysis
- Fast decision making
- Integrated with BI dashboards

6. BENEFITS OF USING KNIME FOR OLAP & REPORTING

- Open source and easy to use
- Scalable workflows
- Supports data integration & transformation
- Supports OLAP, reporting & dashboards
- Powerful analytics & visualization

